Dueling Attention DQN for short-term telescope scheduling (KM40)

Policy and target networks share this architecture. Target values follow double Q-learning; the replay buffer is prioritized by TD error.

Fixed-position state: each target keeps its own 6-dim block, so output i is always target i
Dueling: V(s) is learned once per state while A(s,a) varies per target
Masked actions: illegal observations (outside a window, or infeasible slew/elevation) are never proposed

state $s_t \in \mathbb{R}^{322}$
[t/T, ψ_t, ε_t, n_t/N]
[x_1,t , … , x_N,t]
4 + 6×N = 322

Obs encoder
Linear 4→64 · ReLU
Linear 64→128 · ReLU · LayerNorm
→ e_ψ (128-d)

Target encoder (×N)
Linear 6→64 · ReLU
Linear 64→128 · ReLU · LayerNorm
→ e_1 … e_N (128-d each)

Multi-head attention (8 heads)
query = e_ψ (telescope state)
key = value = [e_1 … e_N]
attn(Q,K,V) = softmax($QK^T/\sqrt{d_k}$)·V

Residual + LayerNorm
h = LayerNorm(e_ψ + MHA(...))
$h \in \mathbb{R}^{128}$

target values use θ^-; $\theta^- \leftarrow \tau\theta + (1-\tau)\theta^-$

Target net
θ^-
soft update τ=0.005

Value stream
Linear 128→128 · ReLU
Linear 128→1
V(s) — state value

Advantage stream
Linear 128→256 · ReLU · Dropout
Linear 256→N
A(s,a) — per-target advantage

Q(s,a) = V + (A − Ā)
dueling combination
action values for a ∈ {1…N}
output i always refers to target i

Action mask + ε-greedy
mask infeasible targets
(window / slew / elevation)
ε: 0.9 → 0.05, restart 0.45

TelescopeEnv (MDP)
visibility windows · slew model
multi-objective reward
53 targets + advance action

action a_t

next state s_t+1, reward r_t